
Automation and Selenium

SESSION 1 – INTRODUCTION TO AUTOMATION TESTING & SELENIUM

1. List 3 repetitive actions you perform while using any app like Instagram, Zomato, or Flipkart that could be automated with a script or tool. For each, briefly explain why automation would help.

Ans:

App	Repetitive Action	How It Can Be Automated	Why Automation Helps
Instagram	Logging into the account	A Selenium script can open Instagram, enter the username and password, and click the Login button automatically.	It saves time, avoids entering credentials repeatedly, and verifies that the login functionality is working correctly after every update.
Zomato	Searching for restaurants	An automation script can open Zomato, enter a restaurant or food name, click Search, and verify	It reduces manual effort, performs the same search quickly, and checks whether the search feature

		that the results are displayed correctly.	works properly every time.
Flipkart	Searching for a product and applying filters	A script can open Flipkart, search for a product (e.g., Mobile), apply filters such as Brand, Price, and Rating, and verify that filtered products are displayed.	It saves time, ensures filters work correctly, and allows repeated testing without manual effort.

2. Open Selenium IDE (install if needed), record a test that automates logging in to any demo website (e.g., <https://opensource-demo.orangehrmlive.com/>), and play it back to verify the steps are automated.

Hint: Use the 'Record' button, perform login steps, then 'Stop Recording' and use 'Play current test case'.

Ans:

Tool Used

- Selenium IDE (Chrome Extension)

Website

<https://opensource-demo.orangehrmlive.com/>

Test Objective

To automate the login process using Selenium IDE and verify that the user is successfully logged into the OrangeHRM dashboard.

Test Steps

Step No.	Action	Expected Result
1	Install Selenium IDE from Chrome Web Store.	Selenium IDE is installed successfully.
2	Open Selenium IDE and click Create a New Project.	New project is created.
3	Click Record a New Test.	Recording starts.
4	Enter the URL: https://opensource-demo.orangehrmlive.com/	OrangeHRM login page opens.
5	Enter Username: Admin	Username is entered.
6	Enter Password: admin123	Password is entered.

7	Click the Login button.	Login request is submitted.
8	Dashboard page appears.	User is logged in successfully.
9	Stop recording.	Test case is saved.
10	Click Play Current Test.	Selenium IDE performs all recorded steps automatically.

Actual Result

- Browser opened automatically.
- Username and password were entered.
- Login button was clicked.
- Dashboard opened successfully.

Status

PASS

3.Download and launch Katalon Studio, then explore its UI to find where you can create a new test case, add test steps, and run tests. Take screenshots of these three areas and submit them.

Ans:

Aim

To explore the Katalon Studio interface.

Software Used

Katalon Studio

Procedure

Step 1: Download Katalon Studio

1. Download Katalon Studio from the official website.
2. Install the software.
3. Launch Katalon Studio.

Step 2: Create a New Test Case

1. Click File → New → Test Case.
2. Enter the test case name.
3. Save the test case.

Screenshot 1: Create New Test Case

Step 3: Add Test Steps

1. Open the created test case.
2. Select Manual View.
3. Click Add.

4. Add actions such as:

- Open Browser
- Navigate to URL
- Set Text
- Click
- Verify Element

Screenshot 2: Add Test Steps

Step 4: Run Test

1. Click the green Run button.
2. Select Chrome browser.
3. Execute the test case.

Screenshot 3: Run Test

Observation

Activity	Observation
Create Test Case	Successfully created
Add Test Steps	Steps added successfully
Run Test	Test executed successfully

Expected Result

The test case should be created, edited, and executed successfully.

Actual Result

All operations were completed successfully.

Status

PASS

Conclusion

Katalon Studio provides a user-friendly interface for creating, editing, and executing automated test cases.

4.Create a comparison table with 4 columns (Tool Name, Type of Testing, Scripting Required, Example Use Case) for Selenium, Cypress, Katalon Studio, and Postman. Fill in the details based on your own research.

Ans:

Tool Name	Type of Testing	Scripting Required	Example Use Case
Selenium	Web Automation Testing	Yes	Login testing, form validation, regression testing
Cypress	End-to-End Web Testing	Yes	Testing modern web applications and user workflows
Katalon Studio	Web, Mobile, Desktop, and API Testing	Optional	Login automation, mobile app testing, regression testing
Postman	API Testing	Optional	Testing REST APIs such as GET, POST, PUT, and DELETE requests

Comparison Summary

Feature	Selenium	Cypress	Katalon Studio	Postman
Open Source	Yes	Yes	Free Version Available	Yes
Web Testing	Yes	Yes	Yes	No
Mobile Testing	No	No	Yes	No
API Testing	Limited	Limited	Yes	Yes
Easy for Beginners	Medium	Easy	Very Easy	Easy

Conclusion

Each automation tool has its own strengths. Selenium is best for web automation, Cypress is suitable for modern web applications, Katalon Studio supports multiple testing types with a user-friendly interface, and Postman is widely used for API testing.

5. Pick one real-world scenario from your daily app usage (such as searching for a product on Flipkart or booking a movie on BookMyShow). Write a short paragraph describing how you would automate this scenario using any automation tool covered in class.

Ans:

Canario

Searching for a mobile phone on Flipkart.

Automation Tool

Selenium WebDriver

Objective

To automate the process of searching for a product and applying filters on Flipkart.

Steps

1. Launch Google Chrome.
2. Open the Flipkart website.
3. Close the login popup if it appears.
4. Enter "Samsung Galaxy S25" in the search box.
5. Click the Search button.
6. Apply filters:
 - Brand: Samsung
 - Price Range: ₹20,000–₹30,000
 - Rating: 4★ & above
7. Sort the products by Popularity.
8. Open the first product from the list.
9. Verify that the product details page is displayed correctly.

Expected Result

The required product should be displayed after applying all filters, and the product details page should open successfully.

Benefits of Automation

- Saves time by eliminating repetitive manual actions.
- Provides consistent and accurate results.
- Reduces human errors.
- Useful for regression testing.
- Can be executed repeatedly without changing the script.

Conclusion

Automating the Flipkart product search process improves efficiency and ensures reliable results. Selenium WebDriver can perform all user actions automatically, making testing faster and more accurate.

Final Conclusion

Automation testing helps testers execute repetitive tasks efficiently, accurately, and quickly. Tools such as Selenium IDE, Katalon Studio, Cypress, and Postman simplify software testing by automating browser actions, application workflows, and API validations. Automation reduces testing time, minimizes human errors, and improves the overall quality of software applications. This assignment demonstrates the basic concepts of automation testing and the practical use of popular automation tools.

